



CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

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I. Project Report

1. Status Report

#	Work Item	Status	Notes (Work Item in Details)
1		Pending	
2		In Progress	
3		Completed	

2. Team Involvements

#	Task	Member	Notes (Task Details, etc.)
1		KienNT	
2		TuanTV	
3		AnhLM	

3. Issues/Suggestions

#	Issue	Status	Notes (Solution, Suggestion, etc.)
1		Pending	
2		In Progress	
3		Completed	

II. Project Management Plan

1. Overall Description

1.1 Test Model

Agile testing breaks away from traditional sequential quality assurance by treating testing as a continuous, parallel activity rather than a distinct phase at the end of development.

Instead of waiting for an entire module to be finished before verification, cross-functional teams build and run tests iteratively within short development cycles (sprints). The model is anchored by two core pillars:

The Test Automation Pyramid: The foundation relies on fast, automated unit and component tests written alongside source code. Mid-tier contract and integration tests continually verify service boundaries, while a targeted set of end-to-end (E2E) automated tests validates full user workflows. Manual and exploratory testing sit at the apex, reserved for complex domain scenarios rather than routine checks.

Continuous Feedback & Shift-Left Validation: Quality checks shift left to the earliest design and coding stages. Developers and QA specialists collaborate on acceptance criteria before code is committed, ensuring that automated regression suites run on every code push via continuous integration (CI) pipelines.

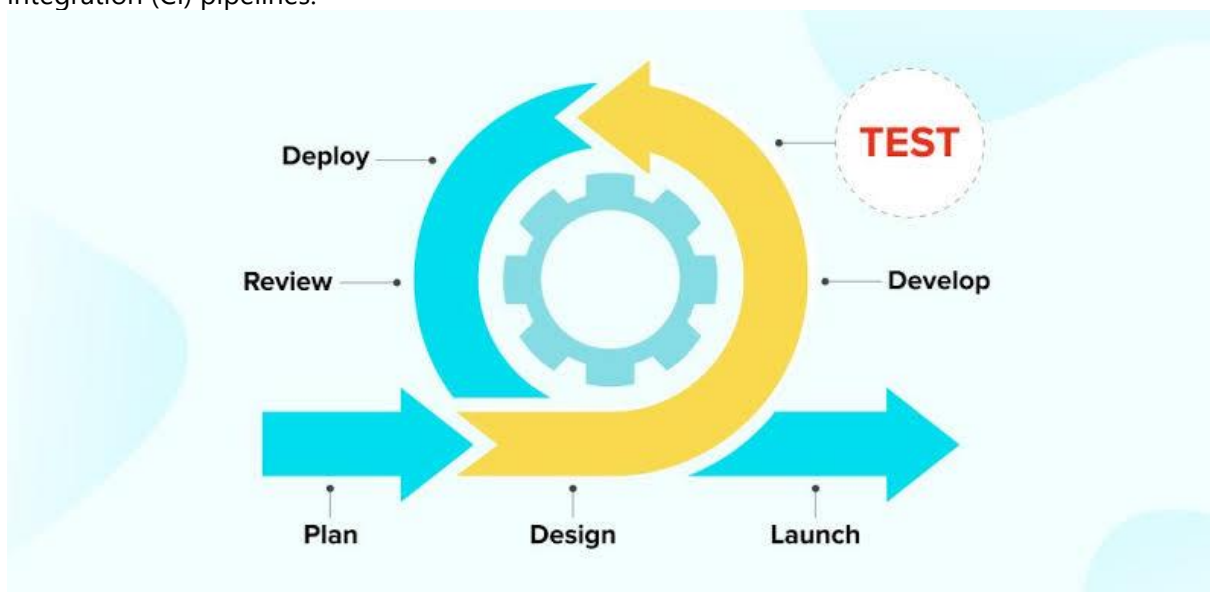


Figure 1. Agile Testing Model

Why we used: S.M.I.L.E is an e-health Dental Practice Management System where patient data integrity, clinical correctness, and payment reliability are non-negotiable. Continuous testing cycles help detect errors as they arise — critical for a healthcare platform where a defect in medical records, prescriptions, or billing can directly harm patients or the clinic's operation. The system is built as a set of microservices (IAM, Clinical EMR, Payment, API Gateway) plus AI services (X-ray analysis, AI booking assistant, KYC OCR) that evolve in parallel across multiple teams; Agile

testing supports quick feedback between teams and flexible adjustment of requirements (e.g., AI model outputs, VNPay integration behavior) throughout the development process.

1.2 Testing Levels

In the software development process, ensuring product quality is a crucial step toward delivering reliable, stable, and user-friendly applications. To achieve this, software must go through several levels of testing, each designed to identify bugs, validate functionality, and confirm that requirements are met. These testing levels range from examining individual components to evaluating the entire system and ultimately checking whether the software meets user expectations. The most commonly applied testing levels include Unit Testing, System Testing, and Acceptance Testing, each serving a distinct purpose.

- **Unit Testing:** The first level of testing, focused on verifying the smallest testable parts of an application, such as individual functions, methods, or classes. In S.M.I.L.E, unit tests are written by developers alongside the code: NestJS services (controllers, services, guards) are tested with Jest. The goal is to ensure that each component operates as expected in isolation, catching bugs early in the development cycle.
- **System Testing:** Performed after all microservices have been integrated behind the API Gateway. At this level, the entire application is tested as a whole to verify that it meets the defined functional and non-functional requirements. It is a type of black-box testing and is carried out by the QA team against the Docker Compose environment. This testing evaluates end-to-end flows that cross service boundaries — such as booking an appointment (Web → Gateway → Clinical EMR → Notification), paying an invoice through VNPay, or uploading an X-ray and receiving AI analysis — as well as usability, performance, security (RBAC, audit logging), and compatibility, ensuring that the system behaves correctly under realistic conditions.
- **Acceptance Testing:** The final level of testing before software is released to production. It is carried out to validate that the software meets business requirements and is ready for delivery to the client or end users. There are typically two types of acceptance testing:
 - **Alpha Testing:** Conducted internally by the development or QA team in a controlled environment, simulating real behaviour of the four user personas (Patient, Dentist, Receptionist, Administrator).
 - **Beta Testing:** Conducted externally by actual users (lecturers and pilot clinic users) in a real-world environment to gather feedback and uncover potential issues not found during internal testing.

Conclusion: A well-structured testing process with clear levels is essential for delivering high-quality software — especially for an e-health system that manages sensitive medical data. By applying Unit Testing, System Testing and Acceptance Testing effectively, the team can detect defects early, validate business needs, and ensure a smooth and safe user experience. These testing levels are not isolated practices but work together to create a robust foundation for successful software deployment.

1.3 Testing Types

- **Functional testing:** is a type of software testing that focuses on verifying that the software's functions operate according to the defined requirements. It checks if each feature of the application behaves as expected with valid and invalid inputs. For S.M.I.L.E this mainly involves user interactions on the web application, REST APIs exposed through

the API Gateway, database state (PostgreSQL), Redis caching, and integrations with external components (Google OAuth, VNPay, mail/OTP delivery, AI inference services). Functional testing typically uses black-box testing techniques and does not require knowledge of the internal code. Its goal is to ensure the correctness and completeness of the system's functionality — from authentication and appointment booking to examination, prescription, imaging, and payment — from the end-user perspective.

- **Non-functional testing:** is a type of software testing that evaluates the performance, reliability, scalability, usability, and security attributes of a system. Unlike functional testing, it does not focus on what the system does, but rather how well it performs under various conditions. For S.M.I.L.E this includes verifying that the AI X-ray analysis responds within acceptable time on CPU inference, that the system handles concurrent booking requests without double-booking treatment rooms or doctors, that patient data is protected by RBAC and audit logging, that VNPay callbacks are secured with checksum and IP validation, and that the interface remains user-friendly for both clinical staff and patients. Non-functional testing helps ensure that the software not only works correctly but also provides a good user experience and meets performance and security expectations required of a healthcare system.

2. Test Plan

2.1 Test Stages

Type of Test/Feature	Stage of Test		
	Unit	System	Acceptance
User Interface test		x	x
Authentication management	x	x	x
Account & user management	x	x	x
Role & permission management	x	x	x
KYC verification management	x	x	x
Clinic & treatment room management	x	x	x
Doctor schedule & work shift management	x	x	x
Patient management	x	x	x
Appointment management	x	x	x
AI booking assistant management	x	x	x
Service catalog management	x	x	x
Examination management	x	x	x
Treatment plan management	x	x	x
Prescription management	x	x	x
Clinical & diagnostic order management	x	x	x
Dental imaging & AI analysis management	x	x	x
Dental chart management	x	x	x
Medical record management	x	x	x
Payment management	x	x	x
Notification management	x	x	x
Audit log management	x	x	
Report & dashboard management	x	x	x

2.2 Resources

a. Human Resources

[List and provide the details on roles and responsibilities of the project members who would involve in testing works, in the table format as below]

Worker/Doer	Role	Specific Responsibilities/Comments
ChinhBC	Leader	Committed to ensuring all testing activities, planning test plans, coordinating the testing team, and progress monitoring for reporting to supervisors.
PhatLT	Tester	Responsible for testing backend Server APIs (IAM, Clinical EMR services) and Web UI.
KhoaDL	Tester	Responsible for testing backend Server APIs (Payment, Gateway services) and Web UI.
NhanTD	Tester	Responsible for testing AI services (X-ray analysis, KYC OCR, booking assistant) and Web UI.

b. Environment

[List and provide the details about the tools (software, hardware, infrastructure) which the project would use for testing. The information can be provided in the table format as below]

Purpose	Tool	Provider	Version
Unit testing (backend services — NestJS)	Jest + @nestjs/testing	OpenJS Foundation / NestJS	30.1.3 / 11.1.6
Unit testing (web frontend — Next.js)	Vitest	VoidZero	3.2.4
Unit testing (AI services — Python)	Pytest	pytest-dev	9.0.3
API testing	Postman	Postman, Inc.	Latest
System testing	Manual	Members of team	—
Acceptance testing	Manual	Lecturer	—

2.3 Test Milestones

Milestone Task	Effort (man day)	Start Date	End Date
Authentication management	4	07/07/2026	10/07/2026
Account & user management	3	07/07/2026	09/07/2026
Role & permission management	2	08/07/2026	09/07/2026
KYC verification management	3	08/07/2026	10/07/2026
Clinic & treatment room management	2	09/07/2026	10/07/2026
Service catalog management	2	09/07/2026	10/07/2026
Doctor schedule & work shift management	3	10/07/2026	12/07/2026
Patient management	3	10/07/2026	12/07/2026
Appointment management	4	10/07/2026	13/07/2026
AI booking assistant management	3	11/07/2026	13/07/2026

Examination management	4	12/07/2026	15/07/2026
Treatment plan management	3	12/07/2026	14/07/2026
Prescription management	2	13/07/2026	14/07/2026
Clinical & diagnostic order management	3	13/07/2026	15/07/2026
Dental imaging & AI analysis management	5	12/07/2026	16/07/2026
Dental chart management	2	14/07/2026	15/07/2026
Medical record management	3	14/07/2026	16/07/2026
Payment management	3	14/07/2026	16/07/2026
Notification management	2	15/07/2026	16/07/2026
Audit log management	1	15/07/2026	15/07/2026
Report & dashboard management	3	15/07/2026	17/07/2026

2.4 Deliverables

[Define delivery of Testing. Refer to the Project Management Plan for more details]

No	Deliverables	Due Date
1	Test plan	17/07/2026
2	Unit test case	17/07/2026
3	System test case	17/07/2026

3. Test Cases

Các test case được mô tả trong hai file bên dưới:

- Unit Test Cases: Report5_Unit Test .xlsx
- Other Test Cases: Report5_ Test Report.xlsx

4. Test Reports

1	Signup	6	0	0	1	4	1	6
2	Login	8	0	0	1	7	0	8
3	Logout	4	0	0	1	3	0	4
4	Login with Google	5	0	0	1	4	0	5
5	Reset Password	5	0	0	1	4	0	5
6	Forgot Password	3	0	0	1	2	0	3
7	Change Password	9	0	0	1	7	1	9
8	View Profile	3	0	0	1	2	0	3
9	Update Profile	3	0	0	1	2	0	3
10	Send OTP	4	0	0	1	3	0	4
11	View Role	2	0	0	1	1	0	2
12	Update Role	3	0	0	1	2	0	3
13	Create Permission	3	0	0	1	2	0	3
14	View Permissions By Role	1	0	0	1	0	0	1
15	Update Role Permissions	7	0	0	1	6	0	7
16	Assign Role	7	0	0	1	6	0	7
17	Revoke Role	5	0	0	1	4	0	5
18	Access Audit Log	1	0	0	1	0	0	1
19	Verify Identity KYC (Phone Number)	3	1	0	1	3	0	4
20	View User List	1	0	0	1	0	0	1
21	Lock/Ban User Account	5	0	0	1	4	0	5
22	Unlock/Unban User Account	3	0	0	1	2	0	3
23	View Clinic Information	1	0	0	1	0	0	1
24	View Clinic Details	3	0	0	1	2	0	3
25	Update Clinic Information	6	0	0	1	5	0	6
26	Add Treatment Room	8	0	0	1	7	0	8
27	View Treatment Room	1	0	0	1	0	0	1
28	Update Treatment Room	3	0	0	1	2	0	3
29	Delete Treatment Room	3	0	0	1	2	0	3
30	Create Work Schedule - On-Call	9	0	0	1	7	1	9
31	Update Work Schedule - On-Call	5	0	0	1	3	1	5
32	View Personal Schedule - Examination	6	0	0	1	5	0	6
33	Register Personal Schedule - Examination	9	0	0	1	7	1	9
34	Update Personal Schedule - Examination	5	0	0	1	3	1	5
35	Notify Schedule Change	5	0	0	1	3	1	5
36	Notify Shift Transfer	3	0	0	1	2	0	3
37	Add Patient Profile	6	0	0	1	5	0	6
38	View Patient Profile	1	0	0	1	0	0	1
39	Update Patient Profile	2	0	0	1	1	0	2
40	Add Medical History	4	0	0	1	3	0	4
41	View Medical Record	3	0	0	1	2	0	3
42	Update Medical Record	2	0	0	1	1	0	2
43	Delete Medical Record	2	0	0	1	1	0	2
44	Add Treatment Profile	7	0	0	1	6	0	7
45	View Treatment Profile	3	0	0	1	2	0	3
46	Update Treatment Profile	2	0	0	1	1	0	2
47	Export Medical Record Profile	6	0	0	1	5	0	6
48	Create Appointment at Facility	24	0	0	1	22	1	24
49	Create Appointment by Specialty	12	0	0	1	10	1	12
50	Create Appointment by Doctor	13	0	0	1	11	1	13
51	Create Appointment Outside Work Hours	14	0	0	1	12	1	14
52	View Appointment	1	0	0	1	0	0	1
53	Edit Appointment	11	0	0	1	9	1	11
54	Cancel Appointment	7	0	0	1	6	0	7
55	Confirm Appointment	1	0	0	1	0	0	1
56	Send Appointment Confirmation	5	0	0	1	4	0	5

57	Send Appointment Reminder	5	0	0	1	4	0	5
58	Initiate Payment	3	0	0	1	2	0	3
59	Confirm Payment / View Payment	1	0	0	1	0	0	1
60	Refund / Cancel Payment	3	0	0	1	2	0	3
61	Chatbot Support for Booking	9	0	0	1	8	0	9
62	View Specialty	3	0	0	1	2	0	3
63	Add Specialty	3	0	0	1	2	0	3
64	Update Specialty	3	0	0	1	2	0	3
65	Delete Specialty	3	0	0	1	2	0	3
66	Enter Symptoms	5	0	0	1	4	0	5
67	View Symptoms	3	0	0	1	2	0	3
68	Edit Symptoms	2	0	0	1	1	0	2
69	Delete Symptoms	2	0	0	1	1	0	2
70	Create Treatment Plan	15	0	0	1	14	0	15
71	View Treatment Plan	3	0	0	1	2	0	3
72	Edit Treatment Plan	3	0	0	1	2	0	3
73	Send Treatment Plan	3	0	0	1	2	0	3
74	Delete Treatment Plan	2	0	0	1	1	0	2
75	Create Electronic Prescription	10	0	0	1	9	0	10
76	Order X-ray/CBCT	10	0	0	1	9	0	10
77	Order Laboratory Test Service	13	0	0	1	12	0	13
78	Order Clinical Test	13	0	0	1	12	0	13
79	Upload Endodontic Image	6	0	0	1	5	0	6
80	Upload X-ray/CBCT Image	6	0	0	1	5	0	6
81	View Dental Image Library	3	0	0	1	2	0	3
82	Attach Image to Treatment Profile	2	0	0	1	1	0	2
83	Edit Image by Treatment Profile	2	0	0	1	1	0	2
84	View Doctor Performance Report	2	0	0	1	1	0	2
85	View Dashboard by Doctor	3	0	0	1	2	0	3
86	View Dashboard by Customer	3	0	0	1	2	0	3
87	Revenue / Financial Report	2	0	0	1	1	0	2
Sub total		430	1	0	87	332	12	431

Test coverage	100.00 %
Test successful coverage	99.77 %
Normal case	20.19 %
Abnormal case	77.03 %
Boundary case	2.78 %

